

Piecewise Functions

Some functions, such as absolute value functions, can be defined in pieces.

The functions in the following examples are piecewise functions.

Definition

A **piecewise function** is a function defined using different rules for different intervals of the domain.

Graphing a Special Piecewise Function

EXAMPLE 1

Graph the piecewise function $f(x) = \begin{cases} -x & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$.

SOLUTION

Create a table of ordered pairs for the first piece in the function.

x	-3	-2	-1	0
$f(x)$	3	2	1	0

The domain of the first piece includes 0 because it is defined for $x \leq 0$.

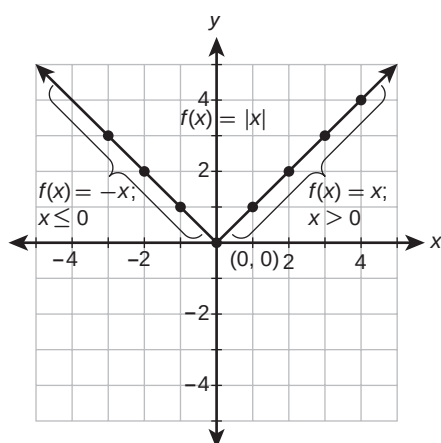
Create a table of ordered pairs for the second piece in the function.

x	1	2	3	4
$f(x)$	1	2	3	4

The domain of the second piece includes all numbers greater than 0.

Plot the points from both tables. Since both pieces of the function are linear, connect the points of each piece with a ray, starting at $(0, 0)$.

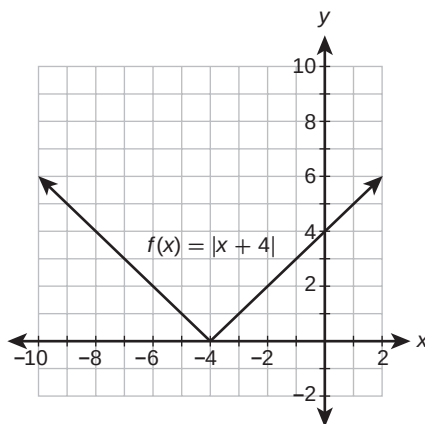
Together, the pieces of the graph form the graph of the absolute value function, $f(x) = |x|$.



Writing an Absolute Value Function as a Piecewise Function

EXAMPLE 2

Write the absolute value function $f(x) = |x + 4|$ as a piecewise function.



SOLUTION

There are two pieces.

The piece on the left is a ray with a boundary point at $(-4, 0)$. Study the table of ordered pairs for points of that piece and determine a rule.

Each range value is 4 subtracted from the opposite of the domain value:

$$f(x) = -x - 4.$$

x	-7	-6	-5	-4
$f(x)$	3	2	1	0

The piece on the right is also a ray with a boundary point at $(-4, 0)$. Study the table of ordered pairs for points of that piece and determine a rule.

Each range value is 4 added to the domain value: $f(x) = x + 4$.

x	-4	-3	-2	-1
$f(x)$	0	1	2	3

Put both rules together. Include the boundary point in only one of the pieces.

$$f(x) = \begin{cases} -x - 4 & \text{if } x \leq -4 \\ x + 4 & \text{if } x > -4 \end{cases} \quad \blacksquare$$

Graphing a Piecewise Function

EXAMPLE 3

Graph the function $g(x) = \begin{cases} 2x & \text{if } x < 1 \\ 1 & \text{if } x = 1. \\ x + 3 & \text{if } x > 1 \end{cases}$

SOLUTION

The domain is divided into three intervals.

For the interval $x < 1$, make a table of values. Although 1 is not included in this interval, use 1 as a value in the table. The point that has 1 as its x -coordinate will be a boundary for this section of the graph. Use an open circle when graphing the boundary point.

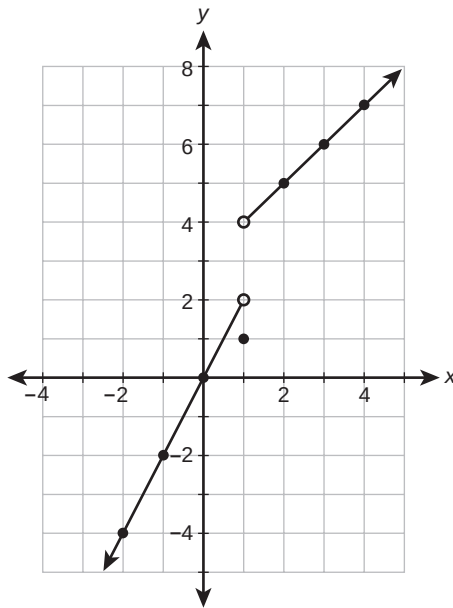
x	$g(x) = 2x$
-2	$2(-2) = -4$
-1	$2(-1) = -2$
0	$2(0) = 0$
1	$2(1) = 2$

For the value $x = 1$, $g(x) = 1$, so graph the point $(1, 1)$.

For the interval $x > 1$, make a table of values. Again, 1 is not included in this interval, but use 1 as a value in the table and use an open circle when graphing the boundary point.

x	$g(x) = x + 3$
1	$1 + 3 = 4$
2	$2 + 3 = 5$
3	$3 + 3 = 6$
4	$4 + 3 = 7$

Graph the function.

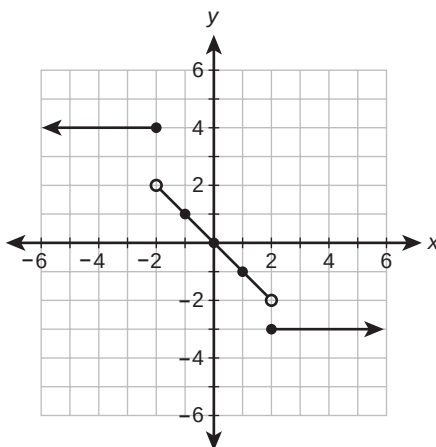


► **Remember** Use open circles for $<$ and $>$. Use closed circles for $\leq$ and $\geq$.

Writing a Rule for a Piecewise Function

EXAMPLE 4

A Write a rule for the graphed piecewise function.



SOLUTION

The left piece is a horizontal ray that extends to the left without end from the point $(-2, 4)$. So it has a constant function value of 4, and the interval of domain values is given by the inequality $x \leq -2$. The rule for this piece is $f(x) = 4$.

The center piece is a line segment that slants down from the point $(-2, 2)$ to the point $(2, -2)$. However, the open circles at $(-2, 2)$ and $(2, -2)$ indicate that these two points are not included in this piece of the graph. Therefore, the interval of domain values is given by the compound inequality $-2 < x < 2$. To find a rule for this piece, create a chart and look for a pattern.

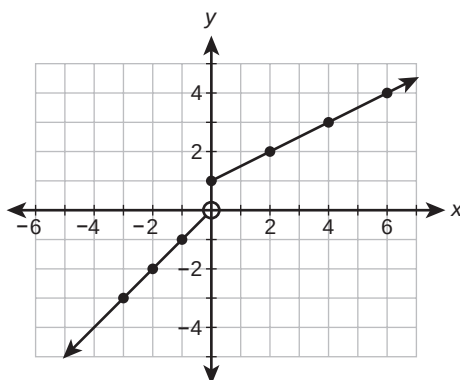
x	-1	0	1
y	1	0	-1

Because y is the opposite of x , the rule is $y = -x$, or $f(x) = -x$.

The right piece is a horizontal ray that extends to the right without end from the point $(2, -3)$. So it has a constant function value of -3 , and the interval of domain values is given by the inequality $x \geq 2$. The rule for this piece is $f(x) = -3$.

Altogether, the rule for the function is $f(x) = \begin{cases} 4 & \text{if } x \leq -2 \\ -x & \text{if } -2 < x < 2 \\ -3 & \text{if } x \geq 2 \end{cases}$.

B Write a rule for the graphed piecewise function.



SOLUTION

The left piece is a ray that slants down to the left without end from the point $(0, 0)$, but it does not include $(0, 0)$. To find a rule for this piece, create a chart and look for a pattern.

x	-1	-2	-3
y	-1	-2	-3

The rule is $y = x$, or $f(x) = x$.

The right piece is a ray that slants up to the right without end from the point $(0, 1)$ and includes $(0, 1)$. Find a rule for this piece.

x	0	2	4	6
y	1	2	3	4

The rule is $y = \frac{1}{2}x + 1$, or $f(x) = \frac{1}{2}x + 1$.

The rule for the function is $f(x) = \begin{cases} x & \text{if } x < 0 \\ \frac{1}{2}x + 1 & \text{if } x \geq 0 \end{cases}$. ■

Finding Function Values of a Piecewise Function

EXAMPLE 5

Find $g(-5)$, $g(-3)$, $g(1)$, and $g(4)$ for the function g .

$$g(x) = \begin{cases} x - 1 & \text{if } x < -3 \\ x^2 - 1 & \text{if } -3 \leq x \leq 3 \\ 2x + 1 & \text{if } x > 3 \end{cases}$$

SOLUTION

Because -5 is in the interval $x < -3$, $g(-5) = -5 - 1 = -6$; -3 is in the interval $-3 \leq x \leq 3$, so $g(-3) = (-3)^2 - 1 = 9 - 1 = 8$; 1 is in the interval $-3 \leq x \leq 3$, so $g(1) = 1^2 - 1 = 1 - 1 = 0$. Since 4 is in the interval $x > 3$, $g(4) = 2 \cdot 4 + 1 = 8 + 1 = 9$. ■